

Chapter 2

Nonparametric Statistics

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Introduction

Nonparametric Statistics: Statistics that do not require assumptions about population parameters or specific distributions (e.g., normality).



When to Use:

- **Small Samples:** Ideal when the sample size is limited.
- **Ordinal Data:** Data is measured on an ordinal scale or in ranks.
- **Unknown Distribution:** When the population distribution is not clearly defined.
- **Median-Focused:** When the median represents the data better than the mean.
- **Unequal Variances:** Used when there is uncertainty regarding variance equality.
- **Independence Testing:** To test the relationship between variables without involving parameter values.



Advantages & Disadvantages



Advantages

1. The calculation procedures are convenient and simple.
2. They can be applied to qualitative data.



Disadvantages

1. In some methods, the actual data used in the analysis are overlooked because they are converted into ranks or symbolic representations.
2. The accuracy of the tests is lower than that of parametric statistics.



Chi-Square Test

This refers to a hypothesis test used with data that are classified into categories, by considering the number (or frequency) of observed values (observed frequency) in each of those categories. Sometimes data that are expressed as numbers or frequencies are referred to as “count data.”

The data used in the chi-square test can be classified into two types as follows:

Types of Chi-Square Data



One-way classified data or “one-factor table”

One-way classified data refers to data organized in a "one-factor table." It represents frequencies (counts) based on a single factor or variable, which is divided into several categories or levels (represented by k).

Example: A table showing the number of students who failed (Grade F) in a Statistics course, categorized by their Faculty (e.g., Architecture, Industrial Education, etc.).

Semester 1/2016

Faculty	Amount
Architecture and Design	15
Industrial Education	5
Industrial Technology and Management	10
Total	30



Types of Tests Used:

1. **Tests of Proportions:** To check if the observed proportions of each group match the theoretically expected proportions.
2. **Goodness-of-Fit Test:** To check if the data distribution follows a specific pattern, such as:
 - Binomial distribution
 - Poisson distribution
 - Normal distribution

The χ^2 -Test (Formula 1) will be used to test whether the proportions in each category conform to the expected values.



Two-way classified data

or “contingency table”

Two-way classified data refers to data that is categorized based on **two different factors** or variables simultaneously. It is usually presented in a **contingency table** (or cross-tabulation), where one factor is represented in rows and the other in columns. This allows researchers to test for an **association or independence** between the two variables.

Example : A table showing the number of students who received an F grade, classified by **both** their "Faculty" (Factor 1) and their "Gender" (Factor 2).

classified by faculty and gender for Semester 1/2016

Faculty	Gender		Total
	Male	Female	
Architecture and Design	10	5	15
Industrial Education	3	2	5
Industrial Technology and Management	2	8	10
Total	15	15	30



The data consist of two factors: Faculty and Gender.

- **The first factor (Faculty)** has 3 levels, denoted as $r = 3$ (where r represents the number of rows).
- **The second factor (Gender)** has 2 levels, denoted as $c = 2$ (where c represents the number of columns).

The statistical tests applied to two-way classified data include:

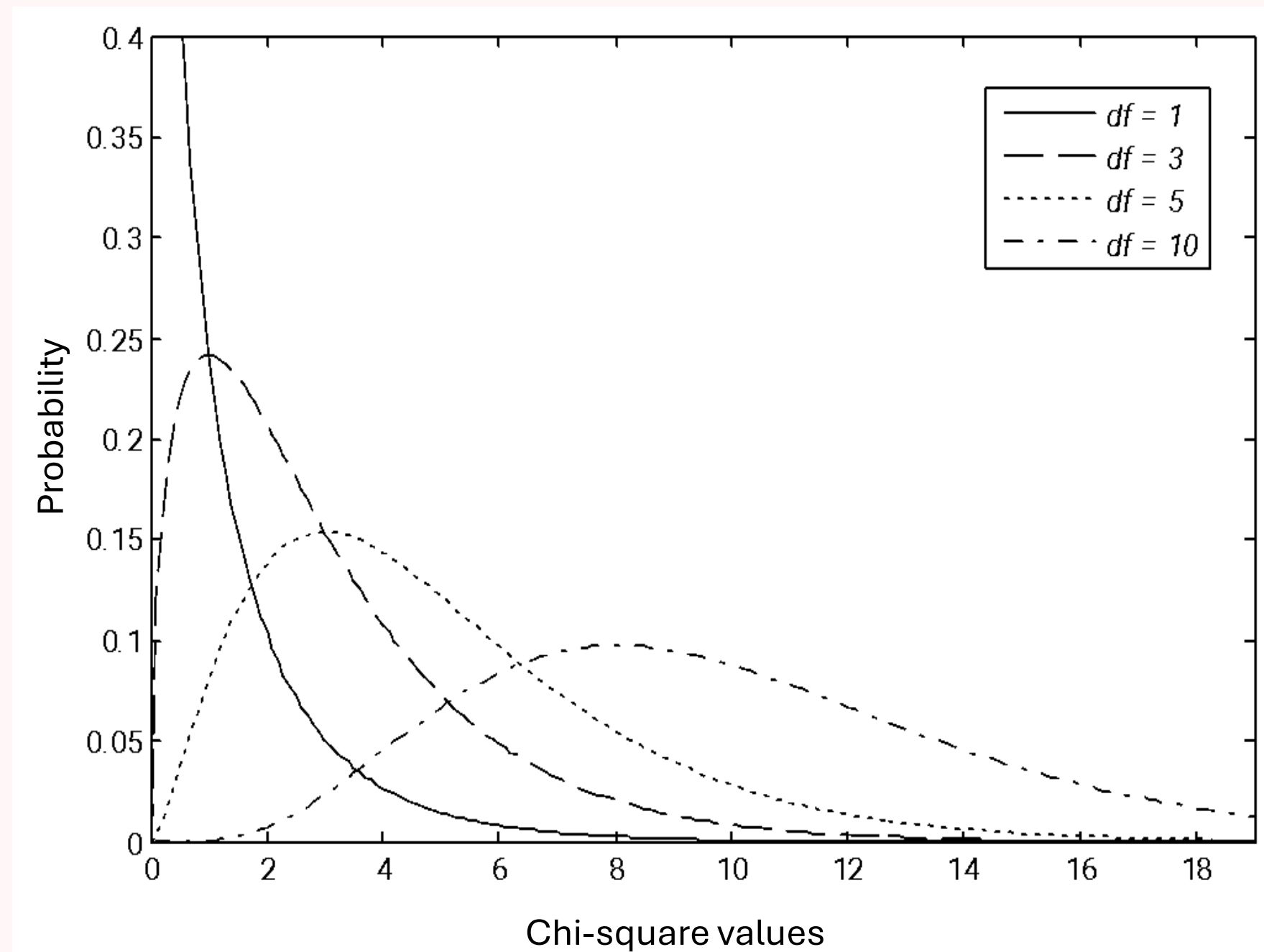
1. **Tests of Independence:** Used to determine whether two specific characteristics or variables in a population are independent of each other.
2. **Tests of Homogeneity:** Used to test whether different populations are similar or consistent in their distribution.

The χ^2 -Test (Formula 2) will be used to test the independence of the two factors.



Properties of the Chi-square Distribution

- 1) A chi-square random variable is a continuous random variable and takes values in the interval $[0, \infty]$.
- 2) A chi-square random variable has a mean equal to the degrees of freedom and a variance equal to twice the degrees of freedom, where degrees of freedom (d.f.) refers to the degree of independence.
- 3) **The distribution curve of a chi-square random variable is right-skewed** and becomes approximately symmetric as the number of degrees of freedom increases.



A figure illustrating the chi-square distribution at various levels of degrees of freedom.

Hypothesis Testing for One-way Classified Data

✦ **Testing whether proportions (probabilities) conform to the specified theoretical values**

Let the characteristic of interest consist of a total of k categories.

Characteristic of interest category i	frequency (amount O_i)	theoretical probability π_i	expected frequency $E_i = n\pi_i$	$\frac{(O_i - E_i)^2}{E_i}$
category 1	O_1	π_1	$E_1 = n\pi_1$	$\frac{(O_1 - E_1)^2}{E_1}$
category 2	O_2	π_2	$E_2 = n\pi_2$	$\frac{(O_2 - E_2)^2}{E_2}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
category k	O_k	π_k	$E_k = n\pi_k$	$\frac{(O_k - E_k)^2}{E_k}$
Total	$\sum_{i=1}^k O_i = n$	$\sum_{j=1}^k \pi_j = 1$	$\sum_{i=1}^k E_i = n = \sum_{i=1}^k O_i$	$\sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$



1) Formulating the hypotheses for the test

$$H_0 : \pi_1 = \pi_{01}, \pi_2 = \pi_{02}, \dots, \pi_k = \pi_{0k}$$

H_1 : There is at least one value that differs from the theoretical value specified in the hypothesis . H_0

Let π_i be the theoretical probability of the occurrence of the characteristic of interest in category

2) Specify the significance level of the test, equal to α ; $\alpha = 1, 2, \dots, k$

3) The test statistic is:

$$\begin{aligned} \chi_{cal}^2 &= \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \\ &= \frac{(O_1 - E_1)^2}{E_1} + \frac{(O_2 - E_2)^2}{E_2} + \dots + \frac{(O_k - E_k)^2}{E_k} \quad (\text{Formula 1}) \end{aligned}$$

Where O_i is the observed frequency in category i ; $i = 1, 2, \dots, k$

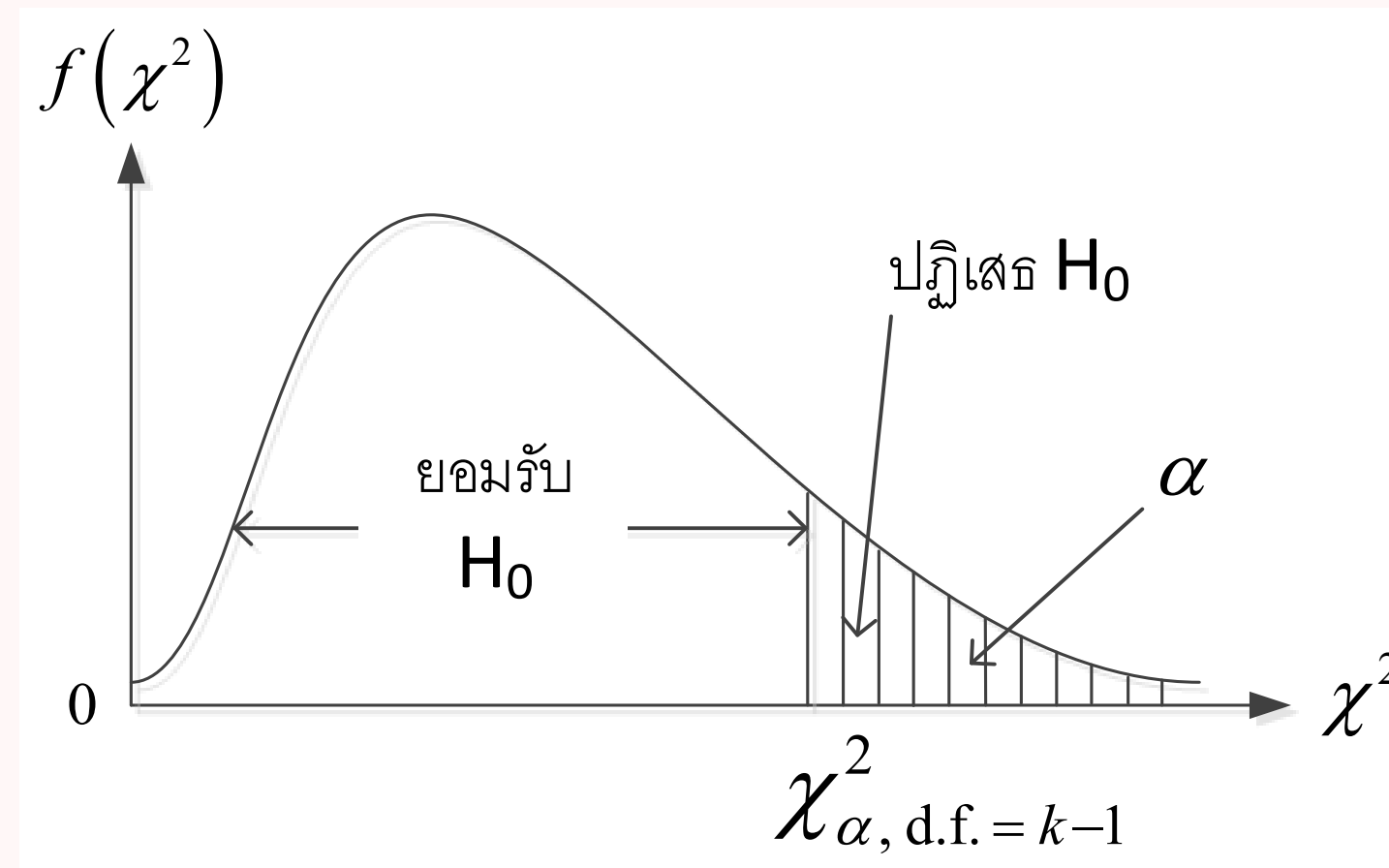
E_i is the expected frequency in category i ; $E_i = n\pi_i$

k is the number of categories or the number of classes.

n is the total frequency or the total sample size obtained. $\sum_{i=1}^k E_i = n = \sum_{i=1}^k O_i$



4. Decision Making



- Reject H_0 and accept H_1 If $\chi^2_{cal} > \chi^2_{\alpha, \text{d.f.} = k-1}$
- Accept H_0 If $\chi^2_{cal} \leq \chi^2_{\alpha, \text{d.f.} = k-1}$

Note: If any class has $E_i = n\pi_i < 5$ the data should be adjusted by combining two adjacent classes (by combining the O_i and E_i values of the two adjacent classes).



Example of testing whether proportions (percentages) conform to the expected theoretical values

Example 2.3 In an experiment of tossing a single die 600 times, it was found that the die showed faces 1, 2, 3, 4, 5, and 6 a total of 70, 80, 64, 120, 136, and 130 times, respectively. We would like to know whether this die is fair at $\alpha = 0.05$ (The value of α is not given directly but must be obtained from the theoretical result of the experiment.)

Solution

Characteristic of interest, category i	frequency (Amount) O_i	theoretical probability π_i	expected frequency $E_i = n\pi_i$	$\frac{(O_i - E_i)^2}{E_i}$
1	$O_1 = 70$	$\pi_1 = \frac{1}{6}$	$E_1 = n\pi_1 = 600 \times \frac{1}{6} = 100$	$\frac{(O_1 - E_1)^2}{E_1} = \frac{(70 - 100)^2}{100} = 9$
2	$O_2 = 80$	$\pi_2 = \frac{1}{6}$	$E_2 = n\pi_2 = 600 \times \frac{1}{6} = 100$	$\frac{(O_2 - E_2)^2}{E_2} = \frac{(80 - 100)^2}{100} = 4$
3	$O_3 = 64$	$\pi_3 = \frac{1}{6}$	$E_3 = n\pi_3 = 600 \times \frac{1}{6} = 100$	$\frac{(O_3 - E_3)^2}{E_3} = \frac{(64 - 100)^2}{100} = 12.96$
4	$O_4 = 120$	$\pi_4 = \frac{1}{6}$	$E_4 = n\pi_4 = 600 \times \frac{1}{6} = 100$	$\frac{(O_4 - E_4)^2}{E_4} = \frac{(120 - 100)^2}{100} = 4$
5	$O_5 = 136$	$\pi_5 = \frac{1}{6}$	$E_5 = n\pi_5 = 600 \times \frac{1}{6} = 100$	$\frac{(O_5 - E_5)^2}{E_5} = \frac{(136 - 100)^2}{100} = 12.96$
6	$O_6 = 130$	$\pi_6 = \frac{1}{6}$	$E_6 = n\pi_6 = 600 \times \frac{1}{6} = 100$	$\frac{(O_6 - E_6)^2}{E_6} = \frac{(130 - 100)^2}{100} = 9$
Total	$\sum_{i=1}^6 O_i = n = 600$	$\sum_{i=1}^6 \pi_i = 1$	$\sum_{i=1}^6 E_i = 600 = \sum_{i=1}^6 O_i$	$\chi^2_{cal} = \sum_{i=1}^6 \frac{(O_i - E_i)^2}{E_i} = 51.92$



1) Formulate the hypotheses for the test

$$H_0 : \pi_1 = 1/6, \pi_2 = 1/6, \pi_3 = 1/6, \pi_4 = 1/6, \pi_5 = 1/6, \pi_6 = 1/6$$

(Fair dice)

H_1 : There is at least one π_i value that differs from the value specified in

$$H_0 ; i = 1, 2, 3, 4, 5, 6$$

(Unfair dice)

2) Specify the significance level of the test, equal to $\alpha = 0.05$



3) The test statistic is: $\chi_{cal}^2 = \sum_{i=1}^6 \frac{(O_i - E_i)^2}{E_i}$

$$= 9 + 4 + 12.96 + 4 + 12.96 + 9$$
$$= 51.92$$

4) The critical region (or the value from the statistical table) is:

Reject H_0 if $\chi_{cal}^2 > \chi_{0.05, k-1=6-1=5}^2 = 11.0705$

5) Decision (accept H_0 or reject H_0): reject H_0 and accept H_1

6) Conclusion: This die is not fair at the 0.05 significance level.



Goodness-of-Fit Test

The Goodness-of-Fit Test is used to assess whether a distribution conforms to the expected distribution. The expected distribution may be a continuous distribution, such as the normal distribution, or a discrete distribution, such as the binomial distribution or the Poisson distribution.

The hypotheses are as follows:

H_0 : The population has a distribution of type Y

H_1 : The population does not have a distribution of type Y

where Y may be normal, binomial, or Poisson.



The test statistic is

$$\chi_{cal}^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$$

When χ^2 = is the symbol for the chi-square statistic

O_i = is the observed frequency

E_i = is the expected (or specified) frequency

n = is the total frequency or the total sample size obtained

π_i = is the probability of occurrence in category i under the expected
distribution



Formula is $E_i = n \times \pi_i$

- The statistic χ^2 follows a chi-square distribution with degrees of freedom

$(k-1)-m$

- m is the number of parameters that must be estimated. Its value depends on the expected distribution being tested. For example, if the hypothesis is

H_0 : the population follows a normal distribution

$m=2$ because the mean and the variance must be estimated $\chi^2 \sim \chi^2_{(k-1-2)}$

- If the hypothesis is

H_0 : the population follows a normal distribution with mean = 5 and variance = 6

- $m=0$ because neither the mean (μ) and the variance (σ^2)

**** Note:** If any class has an expected frequency of less than 5, it should be combined with an adjacent class.



Example 2.6 From 100 water samples collected from a certain river for analysis, to determine the number of living organisms found in each water sample, the data can be summarized as follows.

จำนวนสิ่งมีชีวิต	0	1	2	3	4	5	6	7	รวม
ความถี่	15	30	25	20	5	4	1	0	100

At the 0.05 significance level, test whether the distribution of the number of living organisms follows a Poisson distribution.



Solution

- 1) Formulate the hypotheses for the test

The hypotheses are

H_0 : The distribution of the number of living organisms follows a Poisson distribution.

H_1 : The distribution of the number of living organisms does not follow a Poisson distribution.

- 2) The probability mass function of the Poisson distribution is

$$\pi = f(x) = \frac{e^{-\lambda} \lambda^{x_i}}{x_i!} \quad ; \quad x = 0, 1, 2, \dots$$

- 3) In this case, the parameter λ is unknown, so it is estimated by the sample mean.

$$\bar{x} = \frac{15(0) + 30(1) + 25(2) + 20(3) + 4(5) + 5(4) + 6(1) + 7(0)}{100} = 1.85$$

Number of living organisms (x_i)	O_i	$\pi_i = \frac{e^{-\lambda} \lambda^{x_i}}{x_i!}$	$E_i = n\pi_i$
0	15	0.156	15.567
1	30	0.290	28.955
2	25	0.269	26.928
3	20	0.167	16.696
4	5	0.078	7.763
5	4	0.029	2.888
6	1	0.009	0.895
7	0	0.002	0.238
Total	100	1.0000	

*** Since there are two classes with frequencies less than 5, they are combined with adjacent classes.



Number of living organisms (x_i)	O_i	$E_i = n\pi_i$	$\frac{(O_i - E_i)^2}{E_i}$
0	15	15.567	0.021
1	30	28.955	0.038
2	25	26.928	0.138
3	20	16.696	0.654
4	10	11.785	0.271
Total	100		1.121

4) Specify the significance level of the test $\alpha = 0.05$

The test statistic is:

$$\chi_{cal}^2 = \sum_{i=1}^5 \frac{(O_i - E_i)^2}{E_i} = \frac{(15 - 15.567)^2}{15.567} + \frac{(30 - 28.955)^2}{28.955} + \dots = 4.756$$



- 5) The critical region (or statistics table values) From the χ^2 table, we obtain

$$\chi^2_{\alpha, \text{d.f.}} = \chi^2_{0.05, (5-1)-1=3} = 7.815 \text{ and will reject the hypothesis } H_0 \text{ when } \chi^2_{\text{cal}} > \chi^2_{\alpha, \text{d.f.}}$$

- 6) Decision (accept H_0 or reject H_0): accept H_0

- 7) Conclusion: Since the value of $\chi^2_{\text{cal}} = 1.121 < 7.815$ we therefore accept the main

hypothesis H_0 that is, the distribution of the number of living organisms follows a Poisson

distribution at the 0.05 significance level.

Hypothesis Testing for Two-way Classified Data

Test of Independence of Two Variables

This is a test to determine whether the data in the rows and columns of an $r \times c$ contingency table are related. The data in the horizontal direction are divided into r levels, and the data in the vertical direction are divided into c levels.

Classification B	Classification A				Total
	A_1	A_2	$\dots$	A_c	
B_1	O_{11}	O_{12}	$\dots$	O_{1c}	r_1
B_2	O_{21}	O_{22}	$\dots$	O_{2c}	r_2
$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$
B_r	O_{r1}	O_{r2}	$\dots$	O_{rc}	r_r
Total	c_1	c_2	$\dots$	c_c	n



Where

$$\sum_{i=1}^r \sum_{j=1}^c O_{ij} = \sum_{i=1}^r r_i = \sum_{i=1}^c c_i = n = \sum_{i=1}^r \sum_{j=1}^c E_{ij}$$

And E_{ij} **is the expected number of people** (or the expected frequency) **in row i and column**

Given that r is the total number of rows

c is the total number of columns

O_{ij} is the observed frequency (or number) in row i and column j

r_i is the total frequency in row i ; $i = 1, 2, \dots, r$

c_j is the total frequency in column j ; $j = 1, 2, \dots, c$

n is the total number of observations

π_i is the theoretical probability of occurrence in category i ; $0 \leq \pi_i \leq 1$



Steps in Hypothesis Testing for the Independence of Two Characteristics

- 1) Formulating the hypotheses for the test

H_0 : The two characteristics under study are independent.

H_1 : The two characteristics under study are not independent.

- 2) Specify the significance level of the test, equal to α

$$\begin{aligned}
 3) \quad \text{The test statistic is } \chi_{cal}^2 &= \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \\
 &= \frac{(O_{11} - E_{11})^2}{E_{11}} + \frac{(O_{12} - E_{12})^2}{E_{12}} + \dots + \frac{(O_{rc} - E_{rc})^2}{E_{rc}}
 \end{aligned}$$

where E_{ij} is the expected number of people in row i and column j

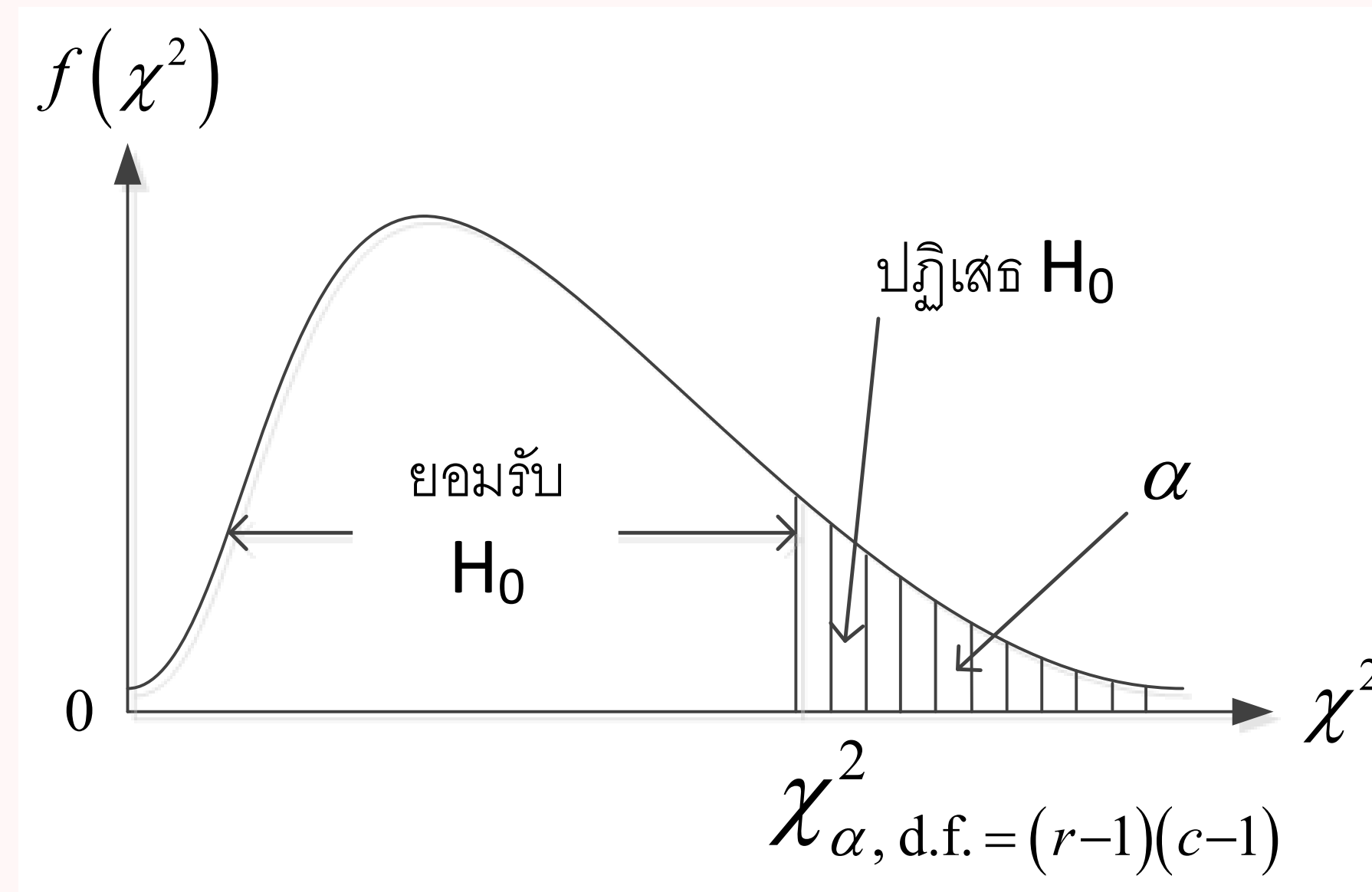
$$E_{ij} = \frac{r_i \times c_j}{n} \quad \text{where } r_i \text{ is the total number of people in row } i$$

c_j is the total number of people in column j

such as $E_{12} = \frac{r_1 \times c_2}{n}$ and $E_{32} = \frac{r_3 \times c_2}{n}$ and so forth.



4) Decision



- reject H_0 and accept H_1 if $\chi^2_{cal} > \chi^2_{\alpha, \text{d.f.}=(r-1)(c-1)}$
- accept H_0 if $\chi^2_{cal} \leq \chi^2_{\alpha, \text{d.f.}=(r-1)(c-1)}$

Example 2.11 A cosmetic company collected data on the gender and work experience of all 500 employees in the company, with the following results:

Gender	Work experience (years)		
	0 – 3	4 – 7	more than 7 years
Male	45	120	85
Female	55	130	65

If there are 150 employees with more than 7 years of work experience, test whether gender and work experience of employees in this company are independent at the 0.01 significance level.



Solution

- 1) Formulating the hypotheses for the test

H_0 : Gender and work experience of the employees are independent.

H_1 : Gender and work experience of the employees are not independent.

- 2) Specify the significance level of the test, equal to

- 3) The test statistic used is
$$\chi_{cal}^2 = \sum_{i=1}^2 \sum_{j=1}^3 \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$
$$= \frac{(45-50)^2}{50} + \frac{(120-125)^2}{125} + \frac{(85-75)^2}{75} + \dots + \frac{(65-75)^2}{75}$$
$$= 0.5 + 0.2 + 1.33 + 0.5 + 0.2 + 1.33$$
$$= 4.06$$



$$E_{11} = \frac{r_1 \times c_1}{n} = \frac{250 \times 100}{500} = 50, \quad E_{12} = \frac{r_1 \times c_2}{n} = \frac{250 \times 250}{500} = 125, \quad E_{13} = \frac{r_1 \times c_3}{n} = \frac{250 \times 150}{500} = 75$$

$$E_{21} = \frac{r_2 \times c_1}{n} = \frac{250 \times 100}{500} = 50, \quad E_{22} = \frac{r_2 \times c_2}{n} = \frac{250 \times 250}{500} = 125, \quad E_{23} = \frac{r_2 \times c_3}{n} = \frac{250 \times 150}{500} = 75$$

4) The critical region (or statistics table values) is: Reject H_0

if $\chi_{cal}^2 > \chi_{0.01, df=(3-1)(2-1)=2}^2 = 9.21034$

5) Decision (accept H_0 or reject H_0): Accept H_0

6) Conclusion: Gender and work experience of the employees are independent at the 0.05 significance level.



Hypothesis Testing concerning the Median of a Single Population: Using the Wilcoxon Signed Method



Ranks Test

- 1) Formulate the hypotheses for the test

$$H_0 : \text{Median} = M_0$$

$$H_1 : \text{Median} \neq M_0$$

Where Median (M_0) is the median of the population (Population Median)

- 2) Specify the significance level of the test, equal to α
- 3) The test statistic is divided into two cases as follows:



3.1) In the case where the sample size $n \leq 20$; the test statistic used is:

$$W = \sum_{i=1}^n R_i^+$$

Where R_i^+ = the rank of observation $|D_i|$ unit i

(after ranking, assign a negative sign to the ranks whose D_i values are negative)

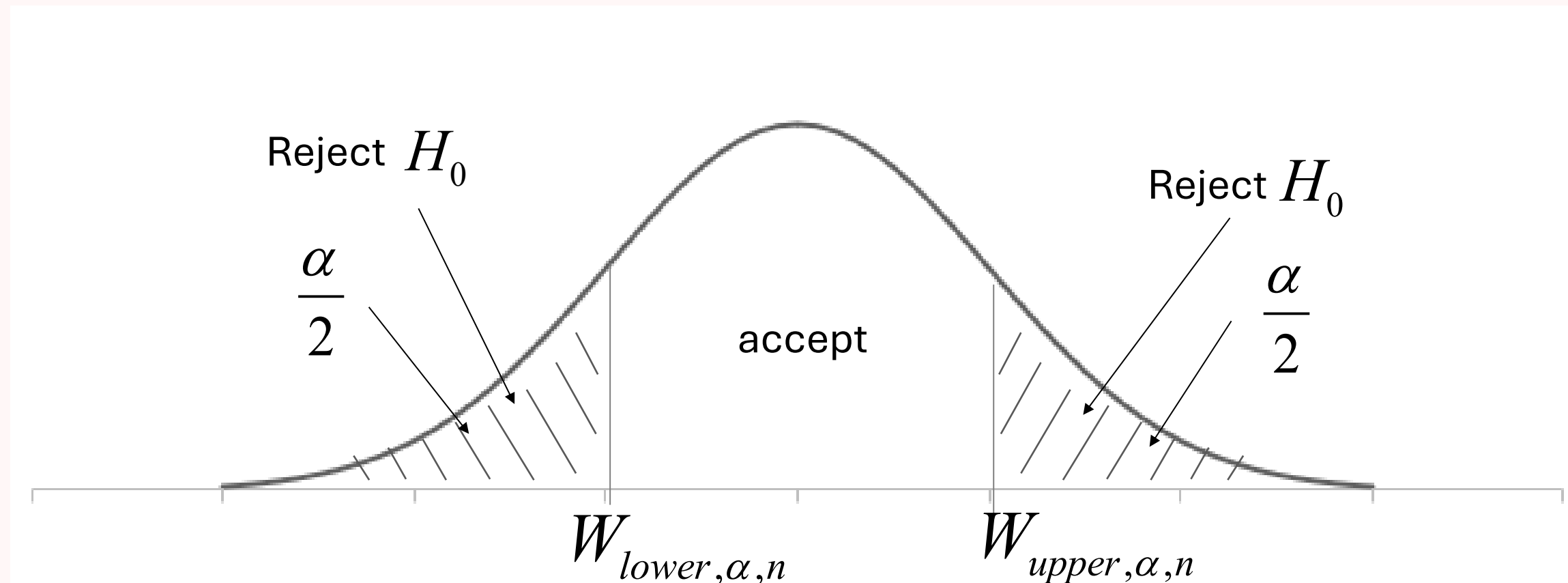
$$D_i = x_i - M_0 ; M_0 \text{ is the median according to} \\ ; i = 1, 2, \dots, n$$

n = the total sample size, excluding the observations for which $D_i = 0$

Note: See the method for calculating on page 27



Decision Making: Two-tailed test



- **Reject** H_0 and **accept** H_1 : if $W \leq W_{\text{Lower}, \alpha, n}$ or if $W \geq W_{\text{Upper}, \alpha, n}$
- **Accept** H_0 : if $W_{\text{Lower}, \alpha, n} < W < W_{\text{Upper}, \alpha, n}$

where $W_{\text{Lower}, \alpha, n}$ and $W_{\text{Upper}, \alpha, n}$ are the two-tailed critical values

obtained from the statistical table “Lower and Upper Critical Values

W of Wilcoxon Signed-Ranks Test”



3.2) In the case where the sample size $n > 20$; the test statistic used is:

$$Z_{Cal} = \frac{W - \mu_W}{\sigma_W}$$

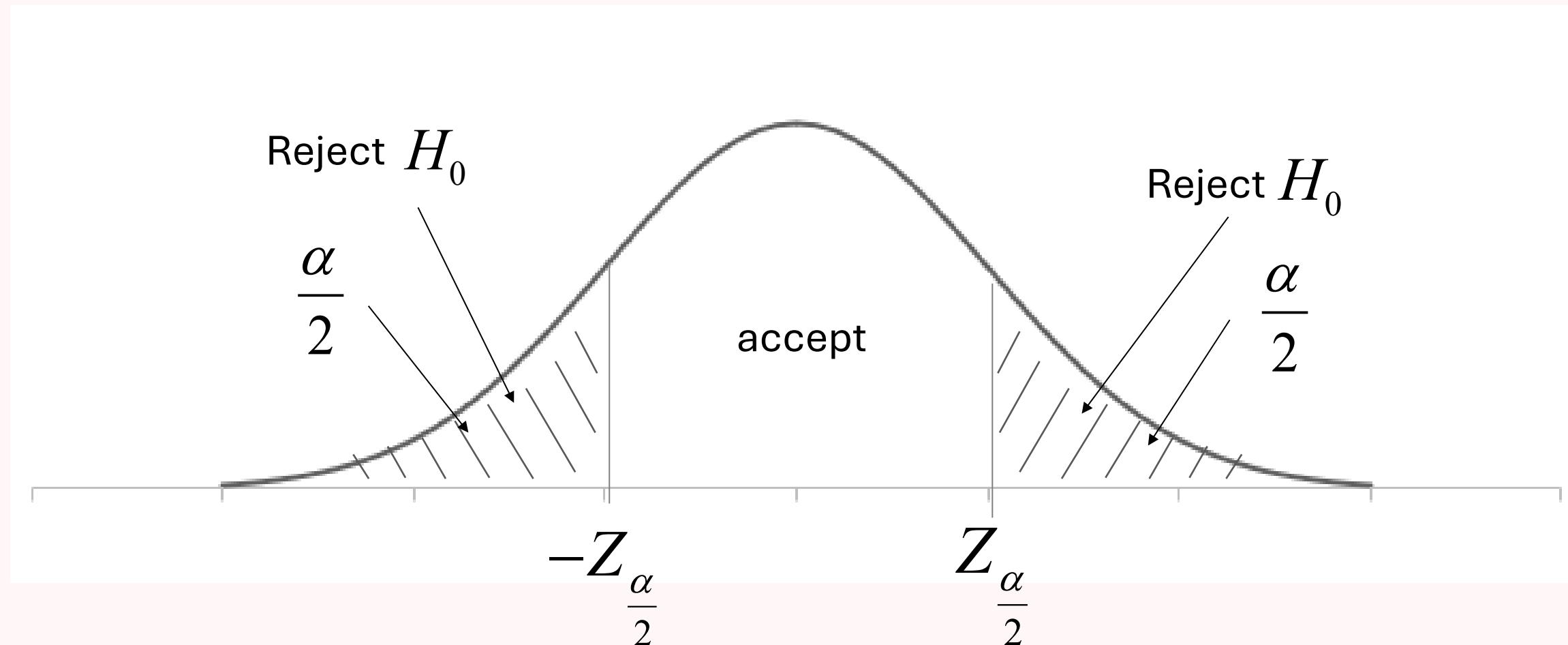
Where $W = \sum_{i=1}^n R_i^+$ (see the method for calculating on page 27)

$$\mu_W = 0$$

$$\sigma_W = \sqrt{\frac{n(n+1)(2n+1)}{6}}$$



Decision Making: Two-tailed test



- Reject H_0 and accept H_1 : if $Z_{cal} < -Z_{\frac{\alpha}{2}}$ and if $Z_{cal} > Z_{\frac{\alpha}{2}}$
- accept H_0 : if $-Z_{\frac{\alpha}{2}} < Z_{cal} < Z_{\frac{\alpha}{2}}$



Steps for calculating W : Case of testing the median of a single population

1. Find the difference between the data value x_i and the median M_0 (M_0 is the value specified in H_0)

Denote this difference by D_i

$$D_i = x_i - M_0; i = 1, 2, \dots, n$$

2. Find the absolute value of D_i denoted by $|D_i|$
3. Arrange the $|D_i|$ values in ascending order and assign ranks to these values.
 - If any $|D_i|$ values are tied, use the method of average ranks (do not assign ranks to observations for which $D_i = 0$)
4. From the ranks obtained in step 3, assign a negative sign to the ranks corresponding to negative D_i values.
5. Compute the total sum of the Positive signed ranks obtained; denote this by W

Example 2.13 An instructor of a statistics course believes that the median score of this examination **does not differ from 50 points. Therefore**, the instructor randomly selected the exam scores of 13 students and obtained the following data:

Student number	exam score (x_i)	$D_i = x_i - M_0$ $= x_i - 50$	$ D_i $	rank of $ D_i $	R_i^+	R_i^-
1	57	$D_1 = 57 - 50 = 7$	7	4	4	
2	70	$D_2 = 70 - 50 = 20$	20	9	9	
3	42	$D_3 = 42 - 50 = -8$	8	5		-5
4	48	$D_4 = 48 - 50 = -2$	2	1		-1
5	77	$D_5 = 77 - 50 = 27$	27	11	11	
6	63	$D_6 = 63 - 50 = 13$	13	7	7	
7	45	$D_7 = 45 - 50 = -5$	5	2.5		-2.5
8	64	$D_8 = 64 - 50 = 14$	14	8	8	
9	55	$D_9 = 55 - 50 = 5$	5	2.5	2.5	
10	39	$D_{10} = 39 - 50 = -11$	11	6		-6
11	73	$D_{11} = 73 - 50 = 23$	23	10	10	
12	78	$D_{12} = 78 - 50 = 28$	28	12	12	
13	50	$D_{13} = 50 - 50 = 0$	0	-	-	-
					63.5	14.5



1. Test the instructor's belief at the 0.05 significance level.

Solution

- 1) Formulate the hypotheses for the test

$$H_0 : \text{Median} = 50 \text{ exam scores}$$

$$H_1 : \text{Median} \neq 50 \text{ exam scores}$$

- 2) Specify the significance level of the test, equal to $\alpha = 0.05$

- 3) The test statistic used is the W-Test, **because $n = 12 < 20$**

$$W = \sum_{i=1}^n R_i^+ = 4 + 9 + \dots + 12 = 63.5$$

- 4) Critical region, that is, reject H_0 if $W \leq W_{\text{Lower},0.05,12}$ or reject H_0 if $W \geq W_{\text{Upper},0.05,12}$

$$W_{\text{Lower},0.05,12} = 13$$

$$W_{\text{Upper},0.05,12} = 65$$

- 5) Decision: Accept H_0



6) Conclusion

Since the test statistic $W = 63.5$ does not fall in the critical region ($13 < 63.5 < 65$) we fail to reject H_0 .

At the 0.05 significance level, there is sufficient evidence to conclude that the median exam score does not differ from 50 points, supporting the instructor's belief.



2. If this instructor additionally selects a further sample of 10 students, making a total of 23 students, and asks for their exam scores, then calculates , and the ranks of in the same way as in (1), the results are: the sum of the Positive Signed Ranks = 150 and the sum of the Negative Signed Ranks = -80. Based on the available data, we would like to know whether this instructor's belief is still correct. Show the method for performing the test at and state the conclusion obtained.

Solution

1) Formulate the hypotheses for the test

$$H_0 : \text{Median} = 50 \text{ exam scores}$$

$$H_1 : \text{Median} \neq 50 \text{ exam scores}$$

2) Specify the significance level of the test, equal to $\alpha = 0.05$



3) The test statistic used is the Z-Test, **because n = 12+10 = 22 > 20**

$$Z_{Cal} = \frac{W - \mu_W}{\sigma_W} = \frac{150 - 0}{61.6} = 2.43$$

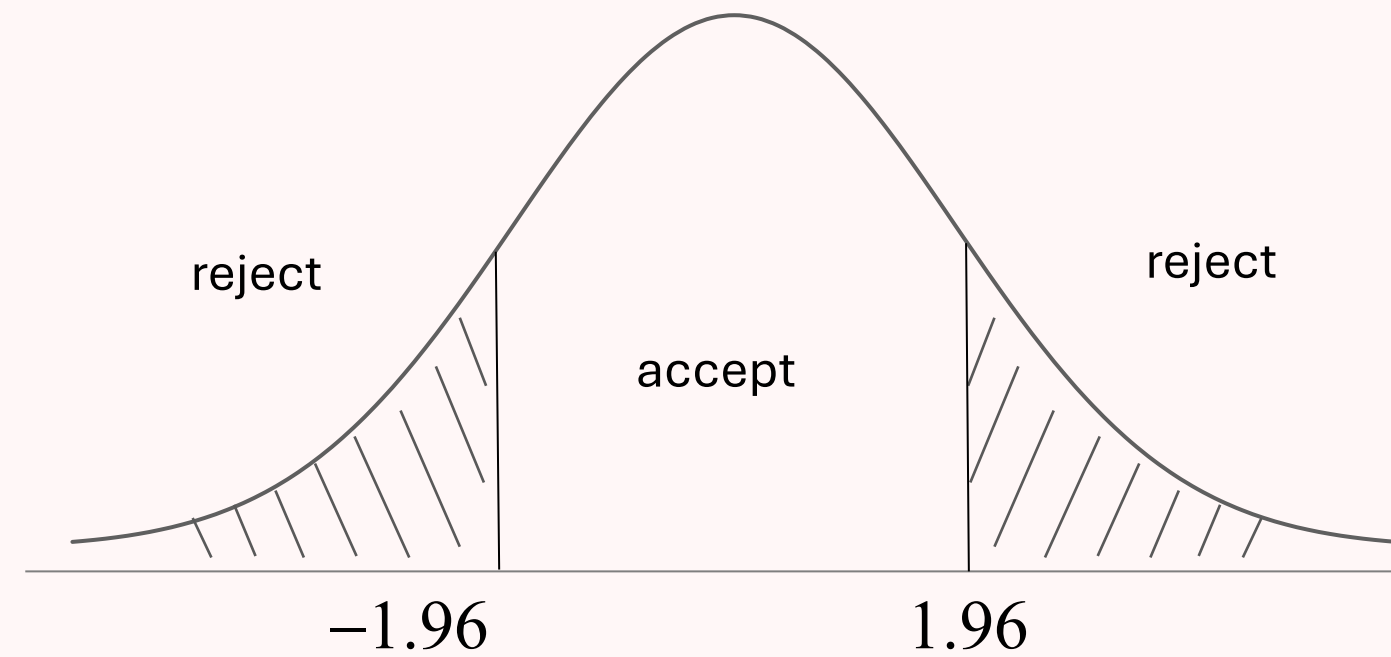
Where $W = \sum_{i=1} R_i^+$

= (**sum** of Positive Signed Ranks)

$$= 150$$

$$\mu_W = 0$$

$$\sigma_W = \sqrt{\frac{n(n+1)(2n+1)}{6}} = \sqrt{\frac{22(22+1)(2(22)+1)}{6}} = 61.6$$



4) Critical region, that is, reject H_0 if $Z_{Cal} < -1.96$ or reject H_0 if $Z_{Cal} > 1.96$ since $Z_{cal} = 2.43 > 1.96$

5) Decision: Reject H_0

6) Conclusion: The median exam score in the statistics course is different from 50 points

at $\alpha = 0.05$.



Hypothesis Testing concerning the Distribution of a Population

Hypothesis Testing concerning Two Related Populations: Using the Wilcoxon Signed Ranks Test with Paired Data

- 1) Formulate the hypotheses for the test

H_0 : The medians of the two populations are not different.

H_1 : The medians of the two populations are different.

- 2) Specify the significance level of the test, equal to α
- 3) The test statistic is divided into two cases as follows:



3.1) In the case where the sample size $n \leq 20$; the test statistic used is

$$W = \text{Smaller} \{W^+, W^-\}$$

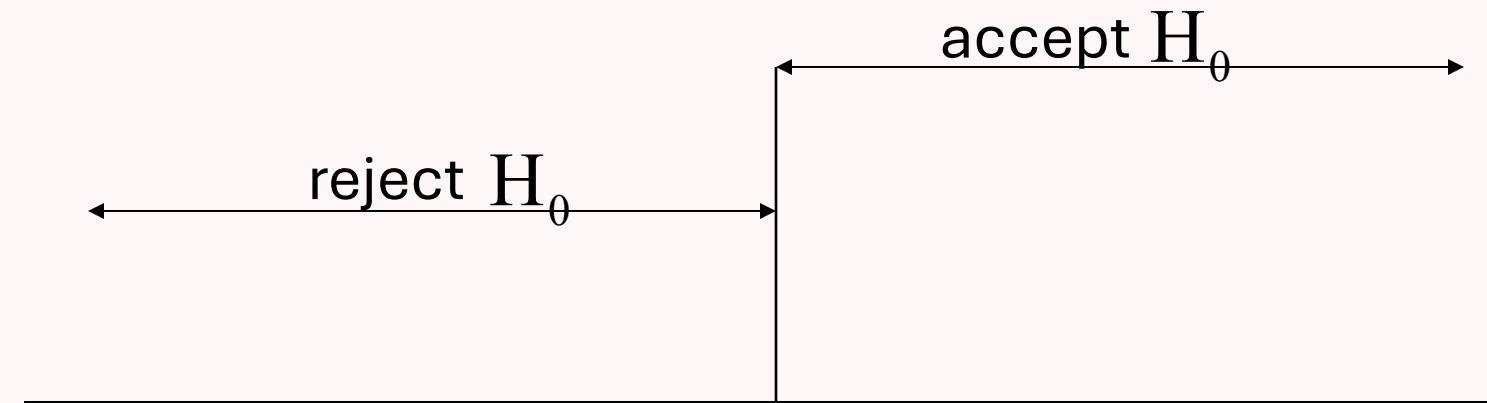
where $W^+ =$ the **sum** of the ranks of observations with **positive** D_i

$W^- =$ the **sum** of the ranks of observations with **negative** D_i

Note: See the method for calculating W on page 31



Decision Making: Two-tailed test



- Reject H_0 and accept H_1 : if $W \leq W_{\alpha,n}$
- accept H_0 : if $W > W_{\alpha,n}$

where $W_{\alpha,n}$ are the two-tailed critical values

obtained from the statistical table “Critical Values for the Wilcoxon Signed-Ranks Test

for $n = 5$ to 50 ”



3.2) In the case where the sample size $n > 20$; the test statistic used is

$$Z_{Cal} = \frac{W - \mu_W}{\sigma_W}$$

Where $W = \text{Smaller}\{W^+, W^-\}$ (see the method for calculating W on page 31)

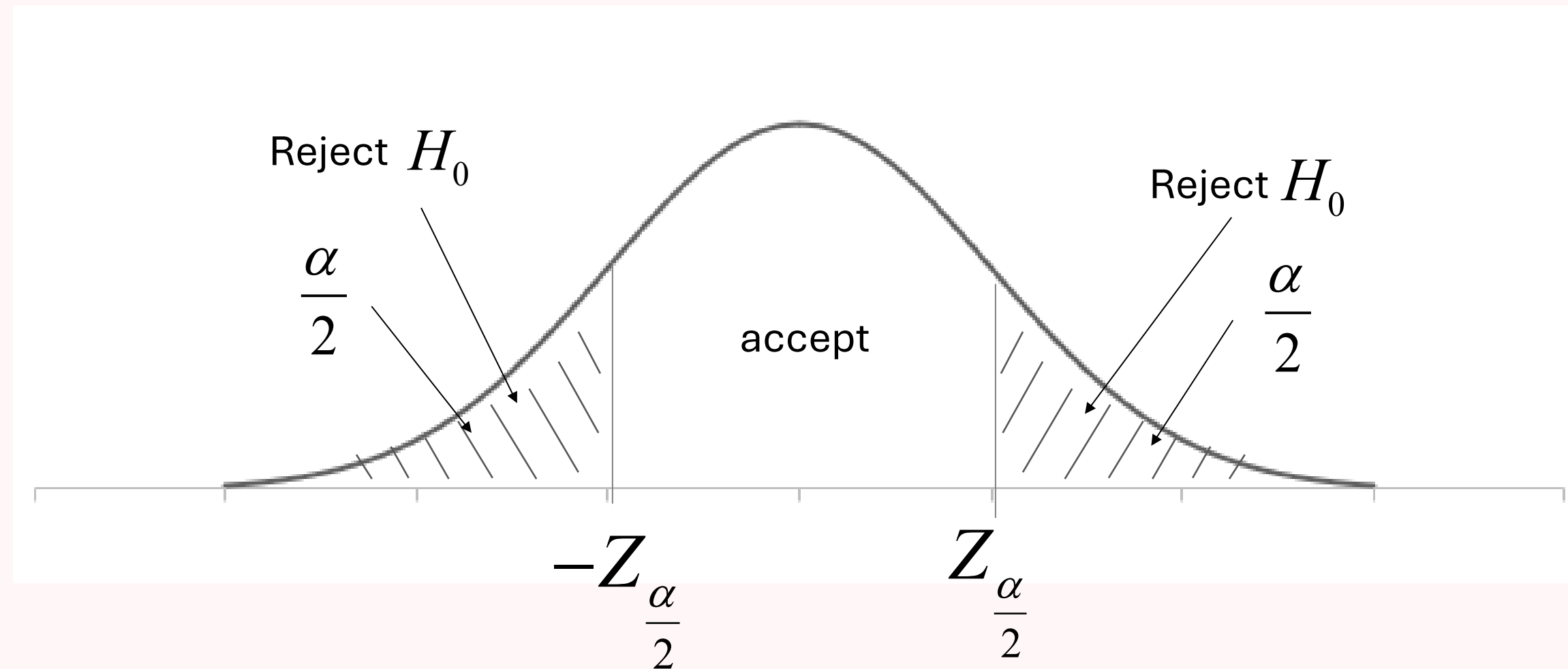
$$\mu_W = \frac{n(n+1)}{4}$$

$$\sigma_W = \sqrt{\frac{n(n+1)(2n+1)}{24}}$$

n = the total sample size excluding the observations for which $D_i = 0$



Decision Making: Two-tailed test



- Reject H_0 and accept H_1 : if $Z_{cal} < -Z_{\frac{\alpha}{2}}$ or if $Z_{cal} > Z_{\frac{\alpha}{2}}$
- accept H_0 : if $-Z_{\frac{\alpha}{2}} < Z_{cal} < Z_{\frac{\alpha}{2}}$



Steps for calculating W : Case of testing two related populations

- 1) Find the difference for each pair of observations; denote this difference by D_i

$$D_i = x_i - y_i ; i = 1, 2, \dots, n$$

where x_i is the observation in group 1 (**before**).

y_i is the observation in group 2 (**after**).

- 2) Find the absolute value of D_i denoted by $|D_i|$
- 3) Arrange the $|D_i|$ values in ascending order and assign ranks to these values.
If any $|D_i|$ values **are tied**, use **the method of average ranks (do not assign ranks to observations for which $D_i = 0$)**
- 4) **Compute the sum** of the ranks corresponding to **positive** $D_i = 0$ denoted by W^+
Compute the sum of the ranks corresponding to **negative** $D_i = 0$ denoted by W^-
- 5) Compute the value $W = \text{Smaller} \{ W^+, W^- \}$

Example 2.15 In an English 2 examination, a pretest was administered before the lesson content, and one week after the lesson the same test was given to the same group of students. From a sample of 26 students, the following data were obtained..

Student No.	Pretest (x_i)	Posttest (y_i)	$D_i = x_i - y_i$	$ D_i $	ตำแหน่ง $ D_i $	W^+	W^-
1	46	76	-30	30	25		25
2	27	36	-9	9	7		7
3	37	53	-16	16	12.5		12.5
4	34	55	-21	21	18		18
5	20	12	8	8	6	6	
6	38	50	-12	12	10		10
7	10	36	-26	26	22		22
8	24	18	6	6	4	4	
9	20	21	-1	1	1		1
10	39	57	-18	18	15		15
11	16	27	-11	11	9		9
12	20	48	-28	28	23		23
13	47	70	-23	23	19		19

Student No.	Pretest (x_i)	Posttest (y_i)	$D_i = x_i - y_i$	$ D_i $	ตำแหน่ง $ D_i $	W^+	W^-
14	45	25	20	20	17	17	
15	40	50	-10	10	8		8
16	46	39	7	7	5	5	
17	32	51	-19	19	16		16
18	49	33	16	16	12.5	12.5	
19	45	69	-24	24	20		20
20	49	52	-3	3	2		2
21	44	60	-16	16	12.5		12.5
22	45	20	25	25	21	21	
23	16	12	4	4	3	3	
24	41	70	-29	29	24		24
25	48	64	-16	16	12.5		12.5
26	35	35	0	0		$W^+ = 68.5$	$W^- = 256.5$

We would like to know whether the medians of the pretest and posttest scores are different at the 0.10 significance level.



Solution

- 1) Formulate the hypotheses for the test

H_0 : The medians of the Pretest and Posttest scores are not different.

H_1 : The medians of the Pretest and Posttest scores are different.

- 2) Specify the significance level of the test, equal to $\alpha = 0.01$
- 3) The test statistic used is the Z-Test

because $n = 26 - 1 = 25 > 20$ (there is one difference value equal to $D_i = 0$, therefore n s reduced)

$$\begin{aligned} Z_{Cal} &= \frac{W - \mu_W}{\sigma_W} \\ &= \frac{68.5 - 162.5}{37.17} \\ &= -2.53 \end{aligned}$$

where $W = \text{Smaller}\{W^+, W^-\} = \text{Smaller}\{68.5, 256.5\} = 68.5$



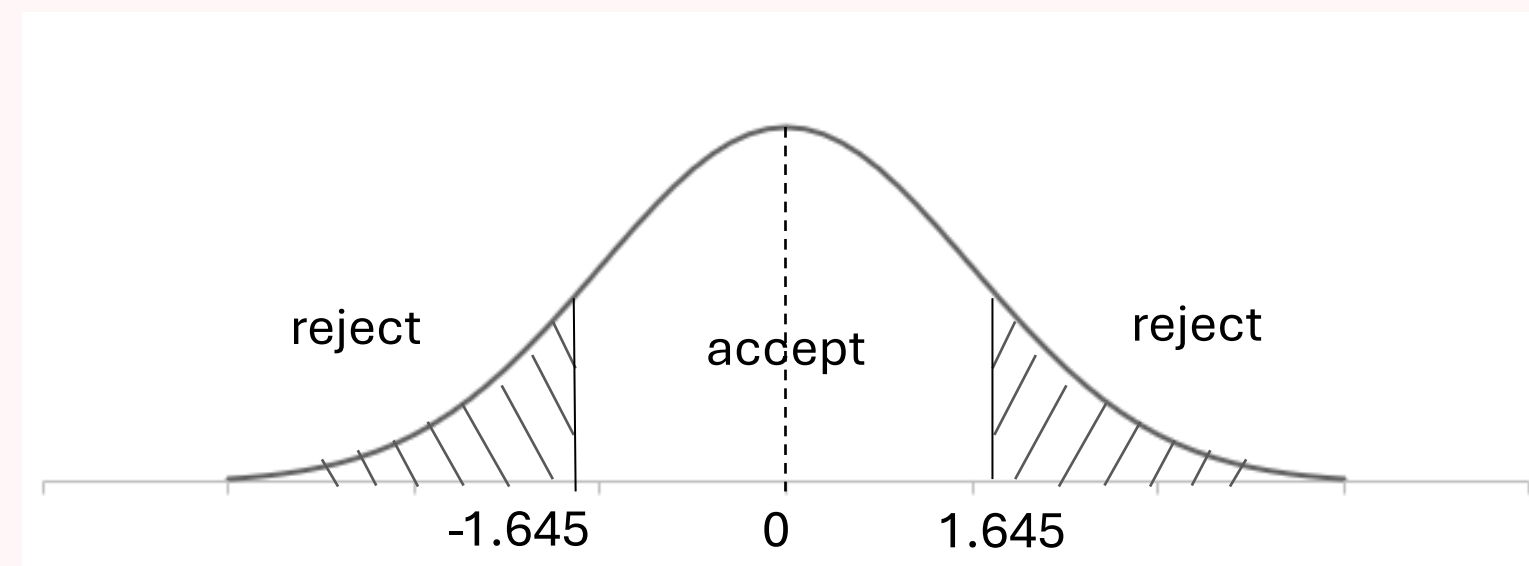
$$\mu_W = \frac{n(n+1)}{4} = \frac{25(25+1)}{4} = 162.5$$

$$\sigma_W = \sqrt{\frac{n(n+1)(2n+1)}{24}} = \sqrt{\frac{25(25+1)(2(25)+1)}{24}} = 37.17$$

4) Critical region, that is, reject H_0 if $Z_{cal} < -1.645$ or reject H_0 if $Z_{cal} > 1.645$

since $Z_{cal} = -2.53 < -1.645$

5) Decision (accept or reject)



6) Conclusion: The medians of the Pretest and Posttest scores are **different** at the 0.10 significance level.



Hypothesis Testing concerning Two Independent Populations: Using the Mann–Whitney U-Test

- 1) Formulate the hypotheses for the test

H_0 : The medians of the two populations are **not different**.

H_1 : The medians of the two populations are **different**.

- 2) Specify the significance level of the test, equal to α
- 3) The test statistic is divided into two cases as follows:



3.1) In the case where the sample sizes are $n_1 \leq 10$ and $n_2 \leq 10$: The test statistic used is

$$U = \text{Smaller}\{U_1, U_2\}$$

Where $U_1 = \sum R_1 - \frac{n_1(n_1+1)}{2}$, $U_2 = \sum R_2 - \frac{n_2(n_2+1)}{2}$

$\sum R_1$ = the **sum** of the ranks of the observations in group 1

$\sum R_2$ = the **sum** of the ranks of the observations in group 2

Note: See the method for calculating on page 37

Decision Making: Two-tailed test

- Reject H_0 and accept H_1 : if $U \leq U_{\alpha, n_1, n_2}$
- Accept H_0 : if $U > U_{\alpha, n_1, n_2}$

Where U_{α, n_1, n_2} are the two-tailed critical values, obtained from the statistical table “Critical Values of U in the Mann-Whitney Test”



3.2) In the case where the sample sizes are $n_1 > 10$ and $n_2 > 10$: the test statistic used is

$$Z_{Cal} = \frac{U - \mu_U}{\sigma_U}$$

Where $U = \text{Smaller}\{U_1, U_2\}$: (see the method for calculating on page 37)

$$\mu_U = \frac{n_1 n_2}{2}$$

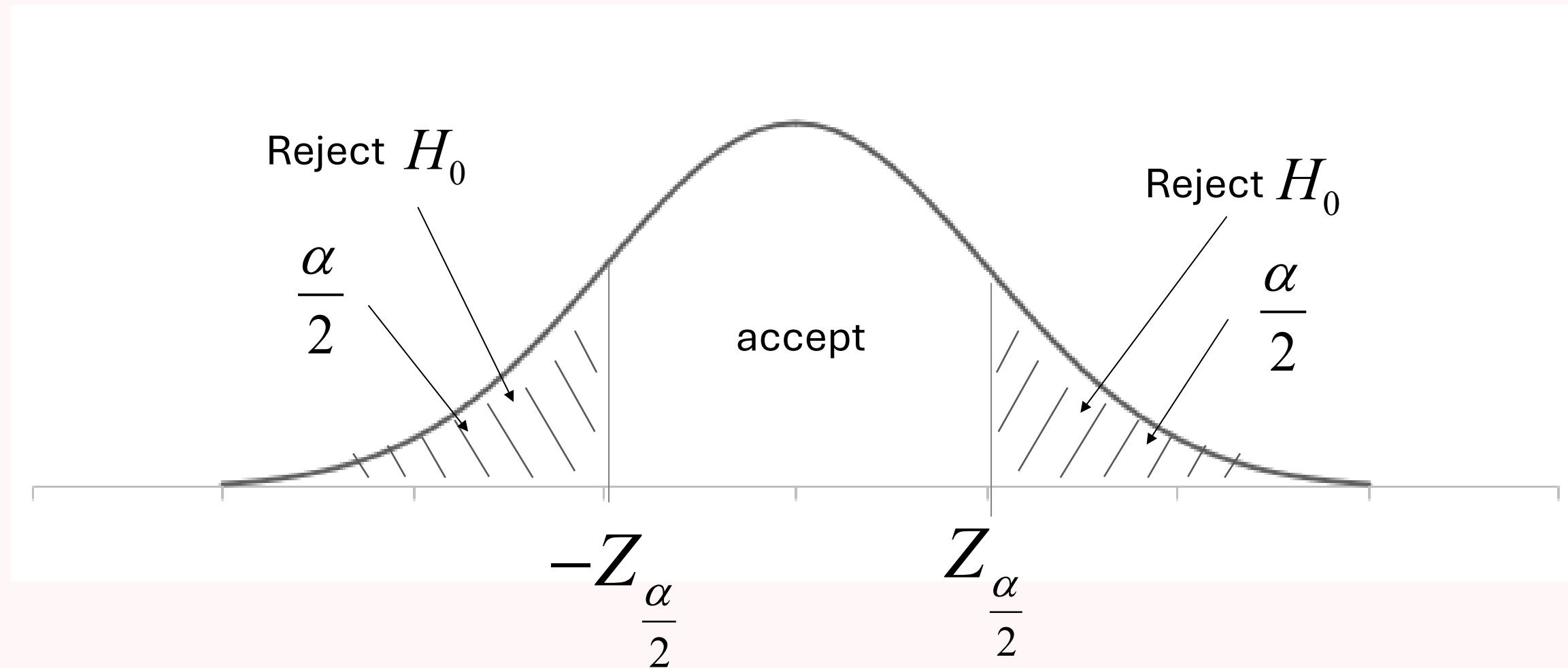
$$\sigma_U = \sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}}$$

n_1 = is the sample size of group 1

n_2 = is the sample size of group 2



Decision Making: Two-tailed test



- Reject H_0 and accept H_1 : if $Z_{cal} < -Z_{\frac{\alpha}{2}}$ or if $Z_{cal} > Z_{\frac{\alpha}{2}}$
- Accept H_0 : if $-Z_{\frac{\alpha}{2}} < Z_{cal} < Z_{\frac{\alpha}{2}}$



Steps for calculating U : Case of testing two independent populations

1. Arrange the data from both groups together in ascending order.
2. Assign ranks to the ordered data.
 - If any values are tied, use the method of average ranks.
3. **Compute the sum of the ranks for group 1**, denoted by $\sum R_1$
Compute the sum of the ranks for group 2, denoted by $\sum R_2$
4. Calculate U_1, U_2 and U

Where $U = \text{Smaller}\{U_1, U_2\}$

$$U_1 = \sum R_1 - \frac{n_1(n_1+1)}{2}$$

$$U_2 = \sum R_2 - \frac{n_2(n_2+1)}{2}$$

and $\sum R_1 =$ **sum** of the ranks of the observations in group 1,

$\sum R_2 =$ **sum** of the ranks of the observations in group 2

$n_1 =$ sample size of group 1

$n_2 =$ sample size of group 2



Example 2.17 An academic advisor wishes to compare the academic performance of 25 advisees, consisting of 12 male students and 13 female students. The advisor collects their GPA data as follows:

male students (group 1)	2.72	3.10	3.75	1.92	2.41	3.72	2.35	3.80	3.25	2.71	3.85	3.90	
female students (group 2)	3.10	2.10	2.62	1.62	2.85	1.95	3.28	1.50	3.71	2.37	3.86	3.88	3.89

We would like to know whether the median academic performance of male students differs from that of female students at the 0.05 significance level.

Solution

Take the data in the above table and assign ranks by arranging the GPAs from smallest to largest and separating them by group as follows.

GPA	rank	group
1.50	1	2
1.62	2	2
1.92	3	1
1.95	4	2
2.10	5	2
2.35	6	1
2.37	7	2
2.41	8	1
2.62	9	2
2.71	10	1
2.72	11	1
2.85	12	2
3.10	13.5	1

GPA	rank	group
3.10	13.5	2
3.25	15	1
3.28	16	2
3.71	17	2
3.72	18	1
3.75	19	1
3.80	20	1
3.85	21	1
3.86	22	2
3.88	23.5	2
3.89	23.5	2
3.90	25	1



1) Formulate the hypotheses for the test

H_0 : The median of the academic performance of male students is not different from that of female students.

H_1 : The median of the academic performance of male students is different from that of female students.

2) Specify the significance level of the test, equal to $\alpha = 0.05$

3) The test statistic used is the Z-test because $n_1 = 12 > 10$ and $n_2 = 13 > 10$

$$Z_{Cal} = \frac{U - \mu_U}{\sigma_U} = \frac{64.5 - 78}{18.38} = -0.73$$

where $U = \text{Smaller}\{U_1, U_2\} = \text{Smaller}\{91.5, 64.5\} = 64.5$

$$U_1 = \sum R_1 - \frac{n_1(n_1+1)}{2} = 169.5 - \frac{12(12+1)}{2} = 91.5$$

$$U_2 = \sum R_2 - \frac{n_2(n_2+1)}{2} = 155.5 - \frac{13(13+1)}{2} = 64.5$$



$$\sum R_1 = \text{The sum of ranks of data in Group 1 (male)}$$

$$= 3 + 6 + 8 + 10 + 11 + 13.5 + 15 + 18 + 19 + 20 + 21 + 25 = 169.5$$

$$\sum R_2 = \text{The sum of ranks of data in Group 2 (female)}$$

$$= 1 + 2 + 4 + 5 + 7 + 9 + 12 + 13.5 + 16 + 17 + 22 + 23 + 24 = 155.5$$

$$\mu_U = \frac{n_1 n_2}{2} = \frac{12 \times 13}{2} = 78$$

$$\sigma_U = \sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}} = \sqrt{\frac{(12 \times 13)(12 + 13 + 1)}{12}} = 18.38$$

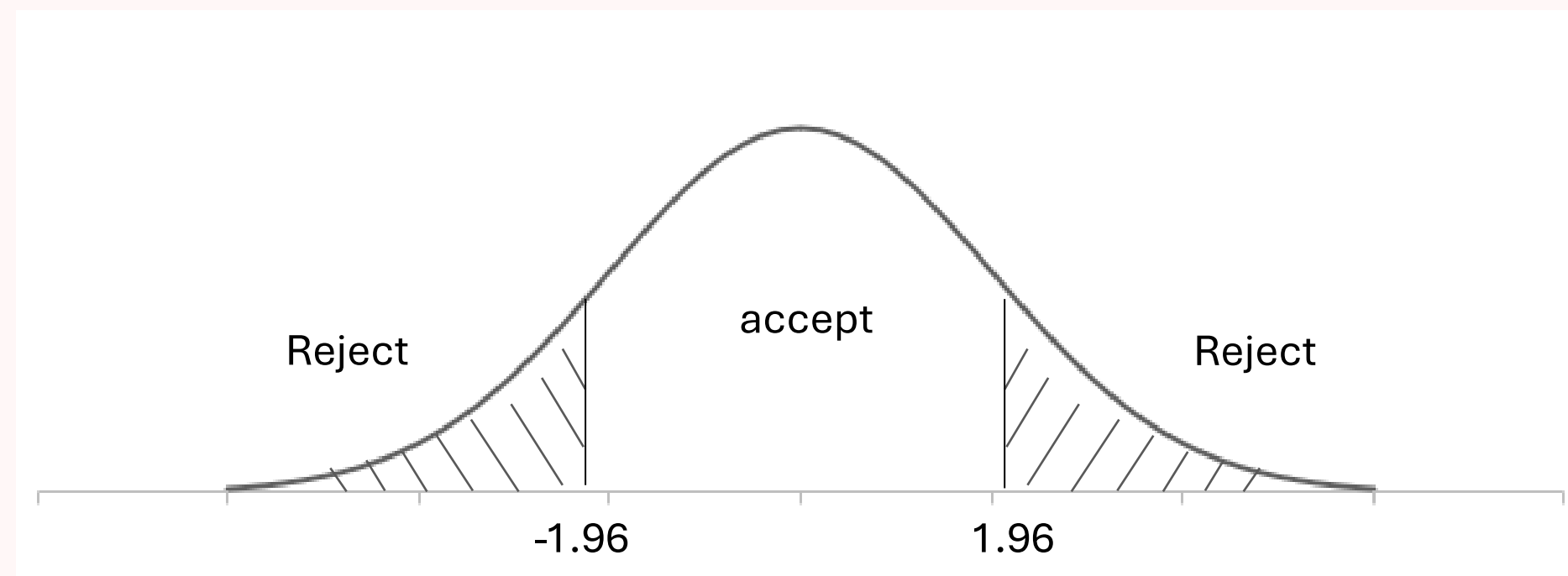


4) Critical region, that is, we will reject H_0 if $Z_{cal} < -1.96$ or reject H_0 if $Z_{cal} > 1.96$

Because $-1.96 < Z_{cal} = -0.73 < 1.96$

5) Decision: Accept H_0

6) Conclusion: The median of the academic performance of male students is not different from that of female students at the significance level of 0.05





Hypothesis testing for populations of 3 or more groups ($k \geq 3$) that are independent of one another: by the Kruskal–Wallis H-Test

- 1) Formulate the hypotheses for the test

$$H_0 : M_1 = M_2 = M_3 = \dots = M_k$$

H_1 : There exists at least one population median that is different from the others.

Where $M_1, M_2, M_3, \dots, M_k$ Is the median in the population of group $1, 2, 3, \dots, k$ respectively.

k = number of groups ; $k \geq 3$

- 2) Specify the significance level of the test, equal to α

- 3) The test statistic used is

$$H = \left[\frac{12}{n(n+1)} \sum_{j=1}^k \frac{R_j^2}{n_j} \right] - 3(n+1)$$

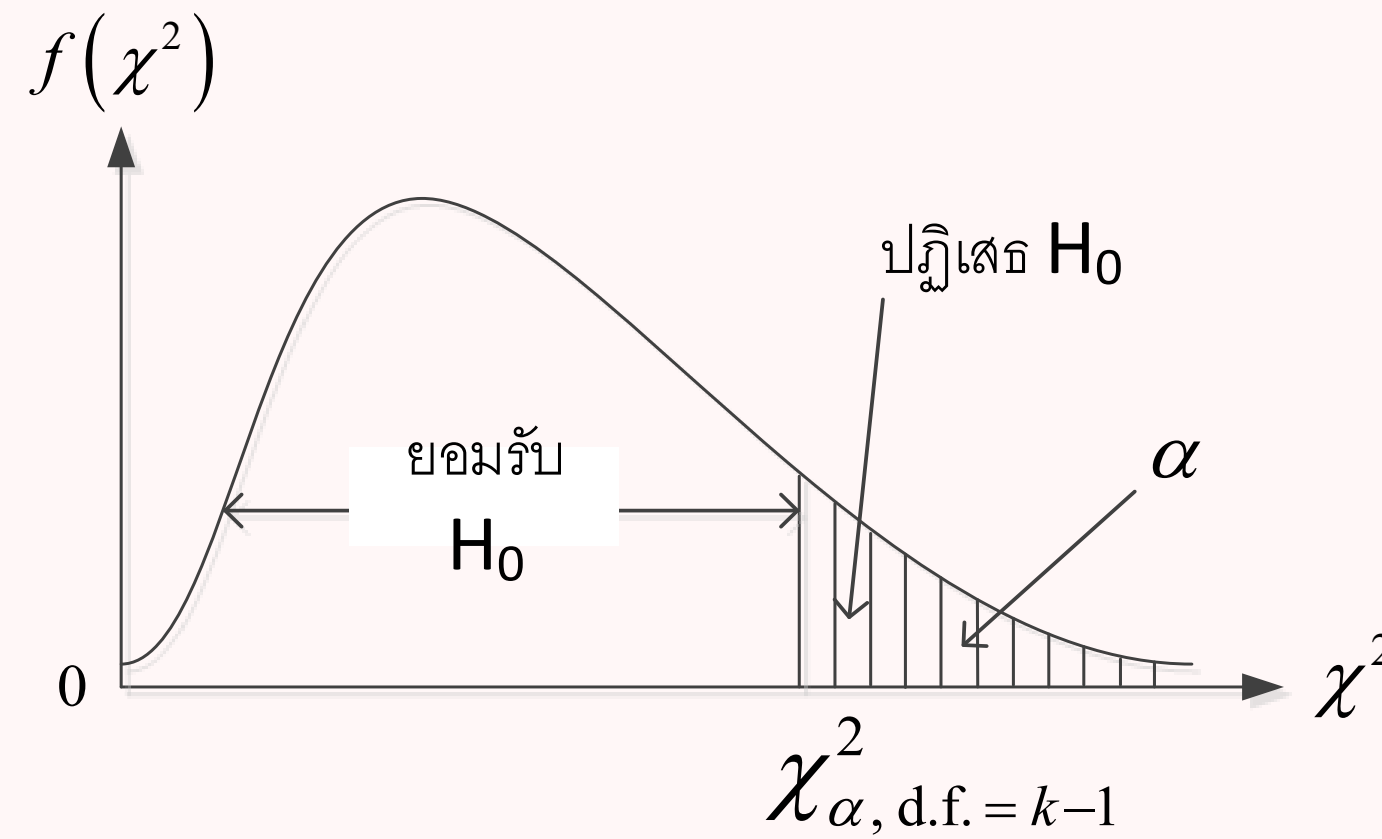
R_j = Where the sum of ranks of the data in group j ; $j = 1, 2, \dots, k$

n_j = the sample size in group j

$n = n_1 + n_2 + \dots + n_k$



4) Decision



- Reject H_0 and accept H_1 if $H > \chi^2_{\alpha, \text{d.f.} = k-1}$

- Accept H_0 if $H \leq \chi^2_{\alpha, \text{d.f.} = k-1}$



Steps for calculating the test statistic H: in the case of testing populations of 3 or more independent groups

1. Arrange the data from all k groups together, from the smallest to the largest, $k \geq 3$
2. **Assign ranks** to the ordered data.
 - If the data contain **tied values**, use the method of assigning **the average rank**.
3.
 - **Find the sum of ranks of Group 1**, denoted by R_1
 - **Find the sum of ranks of Group 2**, denoted by R_2
 - **Find the sum of ranks of Group k**, denoted by R_k
4. Compute the value of H according to the formula on page 42.



Example 2.18 In a study to compare training methods on basic safety in factory buildings, the manager is interested in 4 training methods as follows:

Method 1 Learning by using a manual

Method 3 Watching a video

Method 2 Lecture by an instructor

Method 4 Group discussion

A random sample of 20 workers in the factory was selected. Each worker was randomly assigned to receive one of the four training methods. The training lasted for 3 days. After that, a test was administered to measure the knowledge of the workers who received each of the 4 training methods. The full score on the test is 100 points. The data obtained are as follows:

Method 1	Method 2	Method 3	Method 4
59 (4)	57 (3)	55 (2)	54 (1)
60 (5)	61 (6)	65 (7)	66 (8)
69 (9)	70 (10)	71 (11)	72 (12)
78 (16)	73 (13)	75 (14)	77 (15)
85 (20)	82 (19)	81 (18)	80 (17)
$R_1 = 54$	$R_2 = 51$	$R_3 = 52$	$R_4 = 53$

Determine whether the 4 training methods yield different results. State the conclusion at the 0.01 significance level.



Solution

- 1) Formulate the hypotheses for the test

$$H_0 : M_1 = M_2 = M_3 = M_4$$

(The 4 training methods do not yield different results)

H_1 : At least one population median is different from the others.

- 2) Specify the significance level of the test, equal to $\alpha = 0.01$

- 3) The test statistic used is

$$\begin{aligned} H &= \left[\frac{12}{n(n+1)} \sum_{j=1}^4 \frac{R_j^2}{n_j} \right] - 3(n+1) \\ &= \left[\frac{12}{n(n+1)} \left(\frac{R_1^2}{n_1} + \frac{R_2^2}{n_2} + \frac{R_3^2}{n_3} + \frac{R_4^2}{n_4} \right) \right] - 3(n+1) \\ &= \left[\frac{12}{20(20+1)} \left(\frac{(54)^2}{5} + \frac{(51)^2}{5} + \frac{(52)^2}{5} + \frac{(53)^2}{5} \right) \right] - 3(20+1) \\ H &= \left[\frac{12}{20(21)} \times \left(\frac{11,030}{5} \right) \right] - 63 = 0.02857 \end{aligned}$$

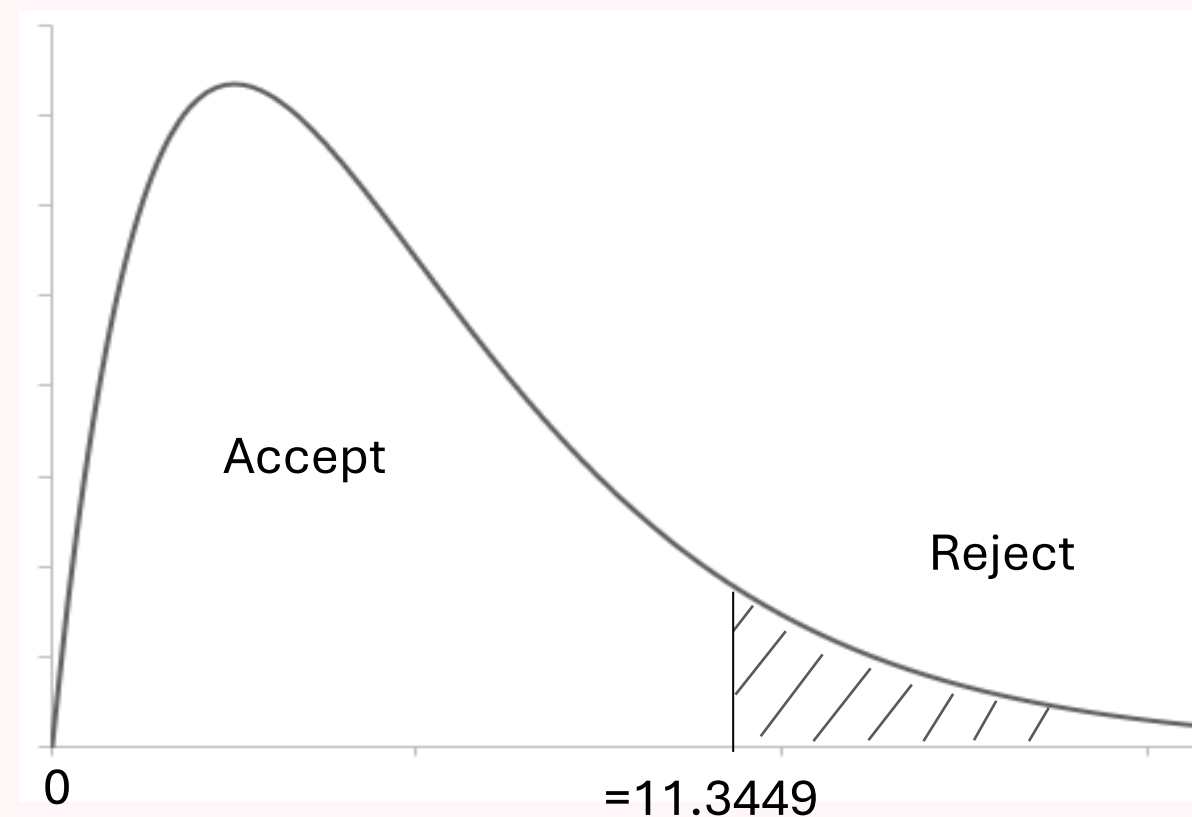


4) Critical region, that is reject H_0 if $H > 11.3449$ and accept H_0 if $H \leq 11.3449$

because $H = 0.02857 < 11.3449$

5) Decision: Accept H_0

6) Conclusion: The 4 training methods do not yield different results at the 0.01 significance level.





Multiple Comparison

It is a method of comparing the population medians pairwise by testing hypotheses to determine which population group causes the rejection of H_0 in the H-Test. In this context, multiple comparisons are performed using Dunn's Procedure.

- **Dunn's Procedure**

It is a method of comparing the population medians pairwise to determine whether they are different, by considering **the absolute value of the difference between the average ranks** for each pair and the Critical Range. Then, observe which pairs have values exceeding this specified Critical Range, and conclude that those population pairs have significantly different medians.



Steps for testing hypotheses concerning population medians pairwise by Dunn's Procedure

- The number of population median pairs that must be tested is

$$= {}^k C_2 = \binom{k}{2} = \frac{k!}{2!(k-2)!}$$

Where k is the number of population groups

- Formulate the hypotheses for the test

$$H_0: M_i = M_j \quad ; \quad i \neq j$$

$$H_1: M_i \neq M_j$$

- Specify the significance level of the test, equal to α

Compute the Critical Range

$$= Z_{\frac{\alpha}{2m}} \sqrt{\frac{n(n+1)}{12} \left(\frac{1}{n_i} + \frac{1}{n_j} \right)}$$

Where $m = \frac{k(k-1)}{2} \quad ; \quad n = n_1 + n_2 + \dots + n_k$

n_i and n_j are the sample sizes of groups i and j , respectively.



- Decision Compare $\left| \bar{R}_i - \bar{R}_j \right|$ with the Critical Range.

- **Reject** H_0 and **accept** H_1 if $\left| \bar{R}_i - \bar{R}_j \right| > \text{Critical Range}$

and conclude that the population medians of that pair are **different**.

- **Accept** H_0 if $\left| \bar{R}_i - \bar{R}_j \right| \leq \text{Critical Range}$

and conclude that the population medians of that pair are **not different**.



Example 2.20 Use Dunn's Procedure to determine which pairs of medians of the diameters of the cylindrical metal rods are different (perform pairwise comparison of medians for 3 pairs).

Solution

In this case, pairwise comparison of medians will be performed for 3 pairs as follows:

I. Comparison of the population medians of Groups 1 and 2

- 1) Formulate the hypotheses for the test

$H_0 : M_1 = M_2$ (The medians of the diameters of the cylindrical metal rods from Machines 1 and 2 are not different.)

$H_1 : M_1 \neq M_2$ (The medians of the diameters of the cylindrical metal rods from Machines 1 and 2 are different.)

- 2) Specify the significance level of the test, equal to $\alpha = 0.05$

- 3) Compute the Critical Range

$$= Z_{\frac{\alpha}{2m}} \sqrt{\frac{n(n+1)}{12} \left(\frac{1}{n_i} + \frac{1}{n_j} \right)} = Z_{\frac{\alpha}{2m}} \sqrt{\frac{n(n+1)}{12} \left(\frac{1}{n_1} + \frac{1}{n_2} \right)} = Z_{\frac{0.05}{2(3)}} \sqrt{\frac{15(15+1)}{12} \left(\frac{1}{5} + \frac{1}{5} \right)} = 2.326(2.83)$$

Therefore, Critical Range = 6.58258

$$\text{Where } m = \frac{k(k-1)}{2} = \frac{3(3-1)}{2} = 3$$



Compare $|\overline{R}_1 - \overline{R}_2|$ with the Critical Range

The average rank of Group 1 is $\overline{R}_1 = \frac{R_1}{n_1} = \frac{65}{5} = 13$

The average rank of Group 2 is $\overline{R}_2 = \frac{R_2}{n_2} = \frac{38}{5} = 7.6$

The average rank of Group 3 is $\overline{R}_3 = \frac{R_3}{n_3} = \frac{17}{5} = 3.4$

Therefore, $|\overline{R}_1 - \overline{R}_2| = |13 - 7.6| = |5.4| < \text{Critical Range} = 6.58$

4) Decision: Accept H_0

5) Conclusion: The medians of the diameters of the cylindrical metal rods from Machines 1 and 2 are not different at the 0.05 significance level.



II. Comparison of the population medians of Groups 1 and 3

- 1) Formulate the hypotheses for the test

$H_0 : M_1 = M_2$ (The medians of the diameters of the cylindrical metal rods from Machines 1 and 3 are not different.)

$H_1 : M_1 \neq M_2$ (The medians of the diameters of the cylindrical metal rods from Machines 1 and 3 are different.)

- 2) Specify the significance level of the test, equal to $\alpha = 0.05$

Compare $|\overline{R}_1 - \overline{R}_3|$ with the Critical Range

Thus $|\overline{R}_1 - \overline{R}_3| = |13 - 3.4| = |9.6| = 9.6 > \text{Critical Range} = 6.58$

- 3) Decision: Reject H_0
- 4) Conclusion: The medians of the diameters of the cylindrical metal rods from Machines 1 and 3 are different at the 0.05 significance level.



III. Comparison of the population medians of Groups 2 and 3

- 1) Formulate the hypotheses for the test

$H_0 : M_1 = M_2$ (The medians of the diameters of the cylindrical metal rods from Machines 2 and 3 are not different.)

$H_1 : M_1 \neq M_2$ (The medians of the diameters of the cylindrical metal rods from Machines 2 and 3 are different.)

- 2) Specify the significance level of the test, equal to $\alpha = 0.05$

Compare $|\overline{R}_2 - \overline{R}_3|$ with the Critical Range

Thus $|\overline{R}_2 - \overline{R}_3| = |7.6 - 3.4| = |4.2| = 4.2 < \text{Critical Range} = 6.58$

- 3) Decision: Accept H_0
- 4) Conclusion: The medians of the diameters of the cylindrical metal rods from Machines 2 and 3 are not different at the 0.05 significance level.



Overall Conclusion

- Arrange the average ranks from the smallest to the largest, together with the group labels.
- From the results of Tests I, II, and III, if any pair of medians is not different, draw a straight line connecting

those groups, as follows:

$$\overline{R}_1 = 13 \quad \overline{R}_2 = 7.6 \quad \overline{R}_3 = 3.4$$

The results can be concluded as follows:

- 1) Machines 1 and 3 have different medians of the diameters.
- 2) Machine 2 has a median diameter that is not different from those of Machines 1 and 3, respectively.



Hypothesis testing concerning the rank correlation coefficient of two variables: by Spearman's Test for Rank Correlation



Rank Correlation Coefficient

Refers to the value used to measure both the magnitude and the direction of the relationship between two variables, by assigning ranks to the data in both groups first and then calculating the rank correlation coefficient.

Let r_s denote the rank correlation coefficient of the sample data by Spearman's method, also called "Spearman's Test for Rank Correlation"

$$r_s = 1 - \frac{6 \sum_{i=1}^n D_i^2}{n(n^2 - 1)}$$

Where $-1 \leq r_s \leq +1$

D_i = the rank of the data - the rank of the data

(D_i = the difference in ranks of variables x and y for pair I)

n = the sample size



Steps for hypothesis testing concerning the rank correlation coefficient

- 1) Formulate the hypotheses for the test

$H_0 : \rho_s = 0$ (The variables x and y are **not** related.)

$H_1 : \rho_s \neq 0$ (The variables x and y are related.)

Where ρ_s is the rank correlation coefficient in the population data by Spearman's method.

- 2) Specify the significance level of the test, equal to α
- 3) The test statistic, divided into 2 cases as follows:

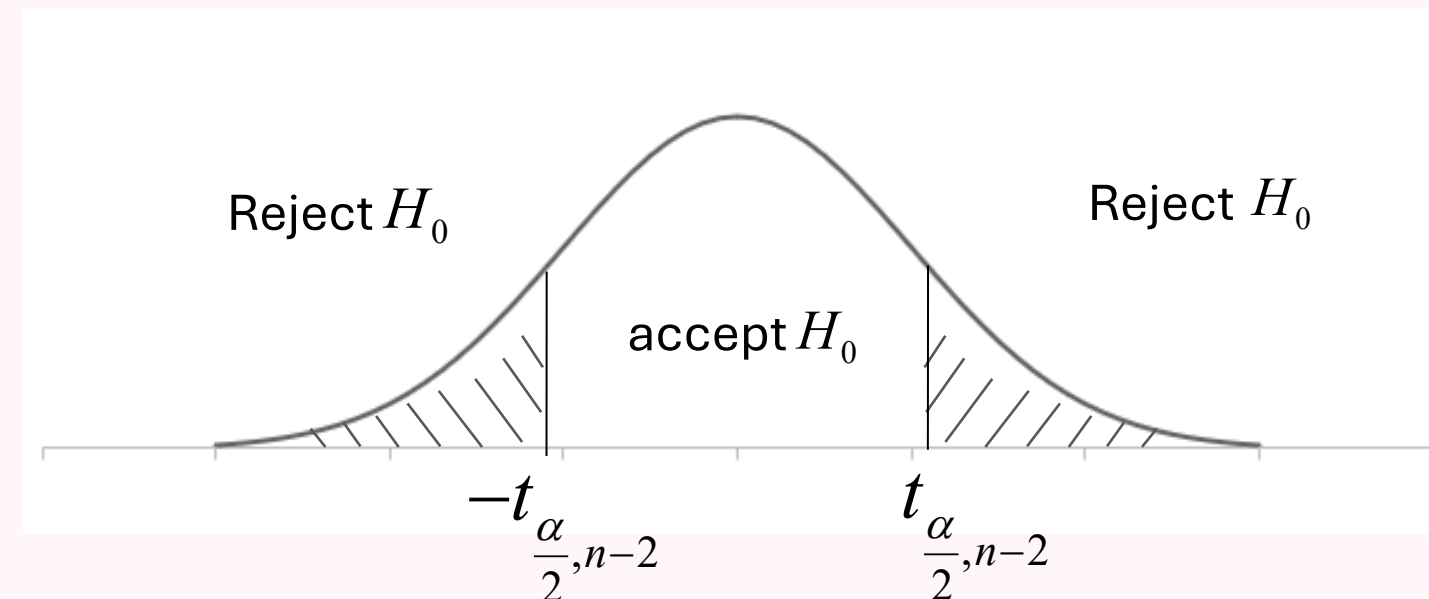


3.1) In the case where the sample size is $n \leq 30$; the test statistic is:

$$t_{cal} = \frac{r_s - \mu_{r_s}}{S_{r_s}} = \frac{r_s - 0}{\sqrt{\frac{1-r_s^2}{n-2}}} = r_s \sqrt{\frac{n-2}{1-r_s^2}}$$

Note: See the method of calculating r_s on page 48

Decision: For the two-tailed test



- Reject H_0 and accept H_1 : if $t_{cal} < -t_{\frac{\alpha}{2}, n-2}$ or if $t_{cal} > t_{\frac{\alpha}{2}, n-2}$
- accept H_0 : if $-t_{\frac{\alpha}{2}, n-2} \leq t_{cal} \leq t_{\frac{\alpha}{2}, n-2}$

Where $t_{\frac{\alpha}{2}, n-2}$ is the statistics table value t that has the right-tail area equal to $\frac{\alpha}{2}$ and has d.f. = $n-2$



3.2) In the case where the sample size is $n > 30$; the test statistic is:

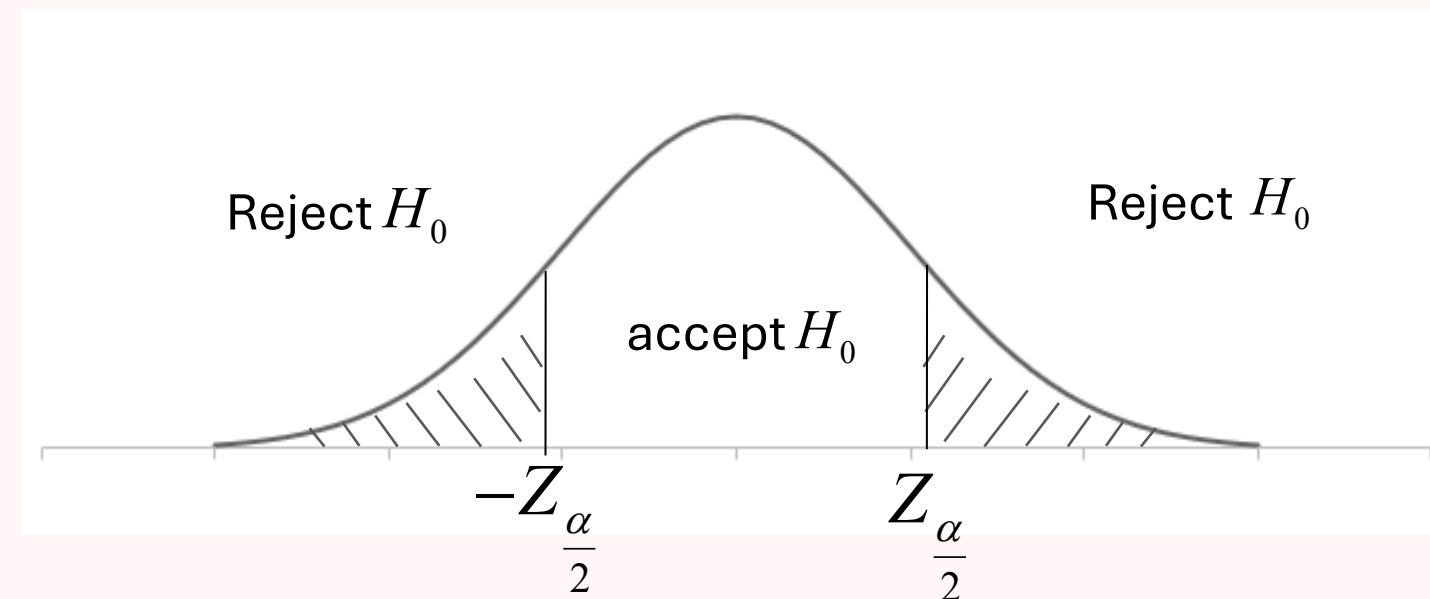
$$Z_{cal} = \frac{r_s - \mu_{r_s}}{\sigma_{r_s}}$$

Where $r_s = 1 - \frac{6 \sum_{i=1}^n D_i^2}{n(n^2 - 1)}$;

$$\mu_{r_s} = 0$$

$$\sigma_{r_s} = \sqrt{\frac{1}{n-1}}$$

Decision: For the two-tailed test



- Reject H_0 and accept H_1 : $Z_{cal} < -Z_{\frac{\alpha}{2}}$ if or if $Z_{cal} > Z_{\frac{\alpha}{2}}$
- Accept H_0 : if $-Z_{\frac{\alpha}{2}} \leq Z_{cal} \leq Z_{\frac{\alpha}{2}}$



Steps for calculating r_s

1. Assign ranks to the data for variables x and y **separately in each group**.
 - If the data contain tied values, use the method of assigning the average rank.
2. Find the difference in ranks of variables x and y for pair i denoted by

The rank of the data x - The rank of the data y D_i

$$D_i = \text{The rank of the data } x_i - \text{The rank of the data } y_i ; i = 1, 2, \dots, n$$

3. Compute the value of r_s using the following formula:

$$r_s = 1 - \frac{6 \sum_{i=1}^n D_i^2}{n(n^2 - 1)}$$

n = the sample size

Example 2.22 In ranking the level of interest of news articles from 5 newspaper brands by two well-known radio news readers, the data obtained are shown in the table. Compute the value of τ and test whether the two news readers have related opinions at $\alpha = 0.05$

	Newspaper					
Newscaster	Thai Rath	Daily News	Khao Sod	Kom Chad Luek	Krungthep Turakij	
Person No. 1 (x_i)	2	3	4	1	5	
Person No. 2 (y_i)	5	2	1	3	4	
D_i (Rank of x_i - Rank of y_i)	-3	1	3	-2	1	
D_i^2	9	1	9	4	1	$\sum D_i^2 = 24$



Solution

- 1) Formulate the hypotheses for the test

$$H_0: \rho_S = 0 \quad (\text{The two news readers do not have related opinions.})$$

$$H_1: \rho_S \neq 0 \quad (\text{The two news readers have related opinions.})$$

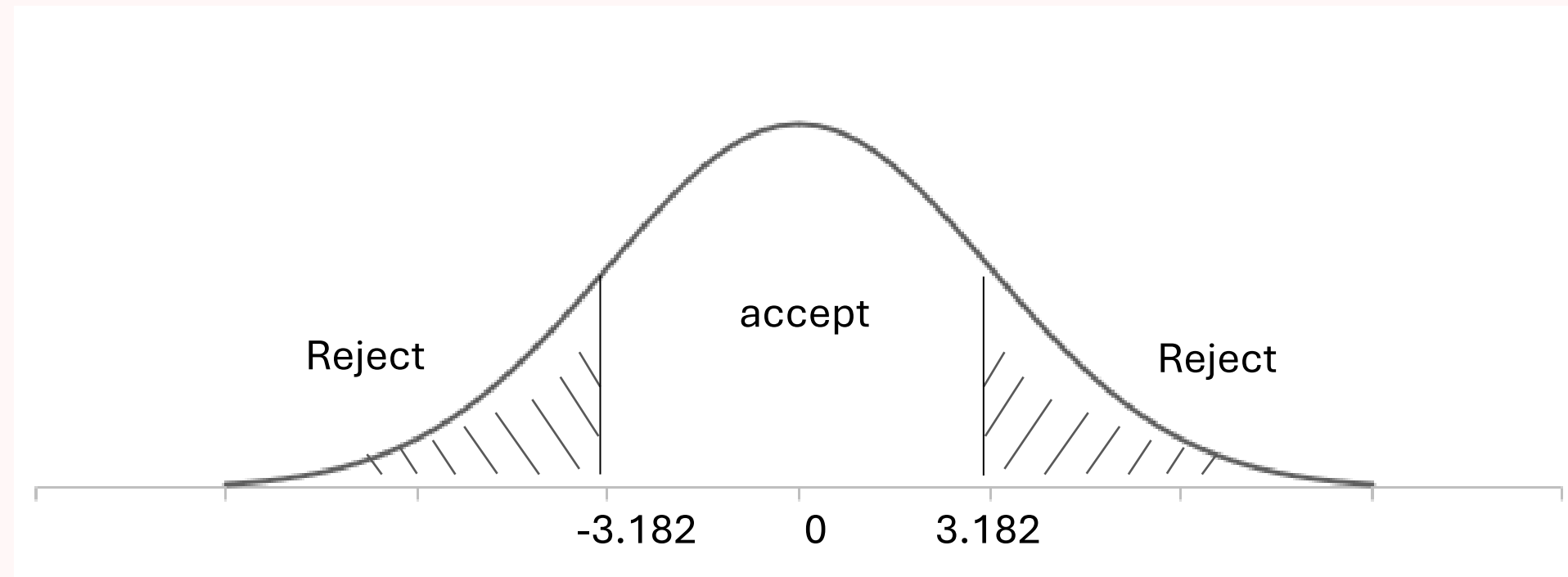
- 2) Specify the significance level of the test, equal to $\alpha = 0.05$

- 3) The test statistic used is $t_{cal} = r_S \sqrt{\frac{n-2}{1-r_S^2}}$ (Since $n = 5 < 30$ the t-test is used.)

$$= (-0.2) \sqrt{\frac{5-2}{1-(-0.2)^2}}$$

$$= -0.35$$

$$\text{where } r_s = 1 - \frac{6 \sum_{i=1}^n D_i^2}{n(n^2 - 1)} = 1 - \left(\frac{6 \times 24}{5(5^2 - 1)} \right) = 1 - 1.2 = -0.2$$



4) Critical region, that is reject H_0 if $t_{cal} < -3.182$ or reject H_0 if $t_{cal} > 3.182$

Because $-3.182 \leq t_{cal} = -0.35 \leq 3.182$

5) Decision: Accept H_0

6) Conclusion: The two news readers do not have related opinions at the 0.05 significance level.

Excercises for Chapter 2

1. From a survey of the number of books borrowed from a certain library during the previous week, the following data were obtained.

	Monday	Tuesday	Wednesday	Thursday	Friday
Number of books borrowed (volumes)	135	108	120	114	146

Test the hypothesis whether the numbers of books borrowed on each day are different or not, using the 0.05 significance level.

2. In order to test whether family size depends on the educational level of the head of the family, a random sample of 200 married men who are old enough to no longer work was selected. It was found that each person with different educational levels has a family with the following numbers of children.

Educational level	Number of children of each person		
	0 – 1	2 - 3	More than 3
Primary education	14	36	30
Secondary education	18	42	20
Higher education	16	14	10

Using the 0.05 significance level, test whether family size depends on the educational level of the head of the family.



3. From an English subject examination that was graded using letter grades, it was found that males received grade A 30%, grade B 25%, and grade C 45%, while females received grade A 35%, grade B 20%, and grade C 45%. Test whether the ability in learning English depends on gender, using the 0.05 significance level..